
MLH
MAJOR LEAGUE HACKING
DEV
DIVERSION 2K27
AKI 秋
HACKS
CALTECH
CIRCLE
HACKTOBERFEST 26
Team Name: Pollution Spotter
Team Leader Name: Sharanya Mondal
PROBLEM STATEMENT
Make visible pond and canal pollution easier to document and report.

PROPOSED SOLUTION

The reporting gap

Residents can see floating waste or discoloured water, but a photo alone lacks a clear reporting path.

Location, reviewed observations and a usable complaint need to stay together.

For residents, student volunteers and local community groups.

One guided reporting flow

01 Capture
Add a photo and confirm its location.

02 Review
Correct the suggested labels and description.

03 Act
Publish a map report; copy a complaint draft.

USER FLOW / WORKING

1 Add a photo

JPEG, PNG or WebP
Up to 3 MB

2 Confirm location

Use a map pin or
optional GPS

3 Get suggestions

Gemini, or a clearly
labeled manual path

4 Review & publish

Edit labels and text;
explicitly confirm

5 View the map

Reviewed details only;
photos stay private

6 Copy complaint

An editable draft;
the user sends it

Human confirmation is required before publication. Manual fallback remains visibly labeled.

TECHNICAL APPROACH

Proposed deployment: one TypeScript app on Vercel

Browser

Next.js · TypeScript · Tailwind
React Leaflet + OpenStreetMap

Server routes

Validate · review · publish
Authenticated owner writes

Gemini vision

Server-side @google/genai
12-second deadline; fallback

Supabase

Postgres + private Storage
Anonymous auth; owned drafts

Public map: reviewed fields only.
Photos, owner IDs and API keys remain private.

INNOVATION & DIFFERENTIATION

Human-reviewed by design

AI proposes visible labels. The reporter corrects them before anything is published.

Privacy at the public map

Photos and owner IDs stay private. Only approved report details appear on the map.

Works when AI does not

A clearly labeled manual path keeps reporting usable if model access fails.

Observation to action

Location, reviewed observations and complaint copy remain connected.

Visible observations do not establish chemical contamination or water safety.

FEASIBILITY, IMPACT & FUTURE

Eight hours. One complete reporting journey.

H1	→	H3	→	H5	→	H7	→	H8
Contracts		Manual slice		Core flow		Freeze		Demo

Build

Four parallel roles: frontend, backend, vision, map & QA.

Impact to validate

Can a resident create a clear reviewed report and copy a useful complaint?

After the event

Density hotspots, moderation and broader field evaluation.

Current status: proposal and planning pack. Implementation follows the event rules.

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